

Lab 1: Verification of Ohm's Law using Resistors in Series

ESC 221 Basic Principles of Electrical Engineering

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I. Introduction, Experimental Method and Hypotheses

A. Introduction

This experiment aims to verify Ohm's Law by analyzing the relationship between voltage, current, and resistance in a series circuit.

B. Experimental Method

In a series circuit with two resistors (R_1 and R_2), a multimeter will be used to measure the voltage and current across a known resistor (R_2) as the input voltage is varied. Additionally, the resistances of the resistors will be determined using the color code and verified with a multimeter to be within the manufacturer's tolerance, using the following formula for percent error:

$$\%_{Err} = \left(\frac{|Measured - Nominal|}{Nominal} \right) \times 100\%$$

C. Hypotheses

It is hypothesized that measured resistance values will fall within the manufacturer's stated tolerance range. For the circuit, it is expected that the current and voltage measurements will follow a linear relationship as described by Ohm's Law, $V = IR$. The predicted voltage across R_2 is calculated using the voltage divider rule.

$$V_{R2} = \frac{R_2}{R_1 + R_2} \cdot V_{in} \quad [V]$$

Applying Ohm's Law, the predicted current through the resistor is:

$$I = \frac{V_{R2}}{R_2} \quad [mA]$$

II. Equipment List

1. Digital Multimeter

Manufacturer: Manufacturer Name

Model: Model Number

2. DC Power Supply

Manufacturer: Manufacturer Name

Model: Model Number

III. Wiring / Schematic Diagram / Procedure

A. Procedure

1. Verifying Resistance

1. Five resistors ($1\text{ k}\Omega$, $2.7\text{ k}\Omega$, $3.3\text{ k}\Omega$, $4.7\text{ k}\Omega$, and $6.8\text{ k}\Omega$) were selected based on their color codes, noting down their nominal resistance values and tolerance.
 2. A multimeter was set to resistance mode with an appropriate range selected to measure the resistance of each resistor ($20\text{ k}\Omega$ range).
 3. Actual resistance values were recorded by connecting the multimeter leads across each resistor.
2. Constructing Circuit A
1. A series circuit was constructed as illustrated in Fig. 1 using a DC power supply and two $1\text{ k}\Omega$ resistors.
 2. Resistor R_1 was connected between Node 1 (positive power supply terminal) and Node 2.
 3. Resistor R_2 was connected between Node 2 and Node 3 (ground).
3. Constructing Circuit B
1. A series circuit was constructed as illustrated in Fig. 2 using a DC power supply and a $1\text{ k}\Omega$ resistor (R_1) and a $3.3\text{ k}\Omega$ resistor (R_2).
 2. Resistor R_1 was connected between Node 1 (positive power supply terminal) and Node 2.
 3. Resistor R_2 was connected between Node 2 and Node 3 (ground).
4. Measurements and Data Collection
1. For each circuit, the input voltage V_{in} was varied from -8 V to $+8\text{ V}$ in increments of 2 V .
 2. For each voltage setting, the voltage across R_2 was measured using a multimeter connected across Node 2 and Node 3, set to voltage mode with an appropriate range (20 V range).
 3. For each voltage setting, the current through the circuit was measured by connecting the multimeter in series with the circuit at Node 2, set to current mode with an appropriate range (20 mA range).

B. Schematic Diagram

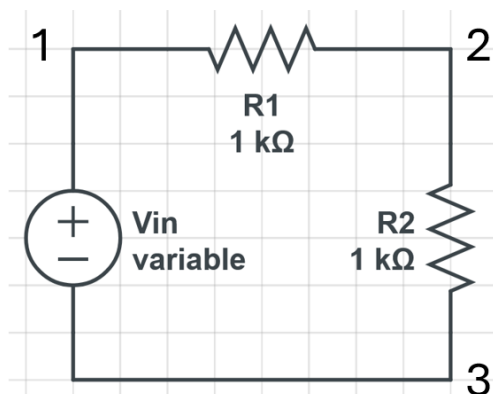


Figure 1: A DC series circuit consisting of a variable voltage source V_{in} and two $1\text{ k}\Omega$ resistors (R_1 and R_2) with labeled nodes 1, 2, and 3 for measurement points.

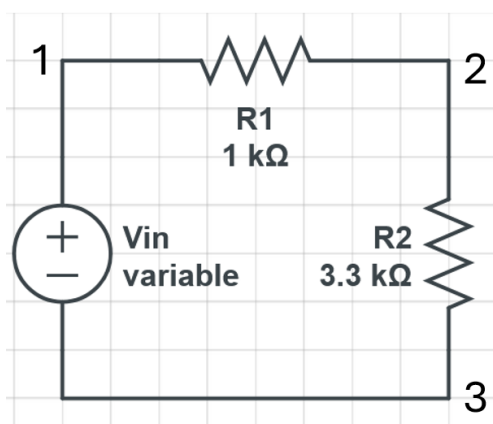


Figure 2: A DC series circuit consisting of a variable voltage source V_{in} and a $1\text{ k}\Omega$ resistor (R_1) and a $3.3\text{ k}\Omega$ resistor (R_2) with labeled nodes 1, 2, and 3 for measurement points.

IV. Results

Table 1: Resistor Tolerance and Measured Values

Resistor ($\text{k}\Omega$)	Nominal Value ($\text{k}\Omega$)	Tolerance (%)	Measured Value ($\text{k}\Omega$)	In Spec?	% Error
1	1	1	0.99	Yes	1.00
2.2	2.2	1	2.18	Yes	0.91
3.3	3.3	5	3.25	Yes	1.52
4.7	4.7	5	4.65	Yes	1.06
6.8	6.8	5	6.72	Yes	1.18

Table 2: Circuit A ($R_2 = 1 \text{ k}\Omega$) Experimental Data

V_{in} (V)	Predicted Current (mA)	Predicted Voltage (V)	Measured Current (mA)	Measured Voltage (V)
2	1.00	1.00	1.00	1.00
4	2.00	2.00	1.96	1.96
6	3.00	3.00	2.94	2.96
8	4.00	4.00	3.98	4.00
-2	-1.00	-1.00	-1.00	-1.00
-4	-2.00	-2.00	-2.03	-1.97
-6	-3.00	-3.00	-2.80	-2.74
-8	-4.00	-4.00	-4.09	-4.00

Table 3: Circuit B ($R_2 = 3.3 \text{ k}\Omega$) Experimental Data

V_{in} (V)	Predicted Current (mA)	Predicted Voltage (V)	Measured Current (mA)	Measured Voltage (V)
2	0.4651	1.5349	0.46	1.53
4	0.9302	3.0698	0.97	3.11
6	1.3953	4.6047	1.42	4.59
8	1.8605	6.1395	1.90	5.92
-2	-0.4651	-1.5349	-0.47	-1.52
-4	-0.9302	-3.0698	-0.95	-3.05
-6	-1.3953	-4.6047	-1.44	-4.62
-8	-1.8605	-6.1395	-1.92	-6.17

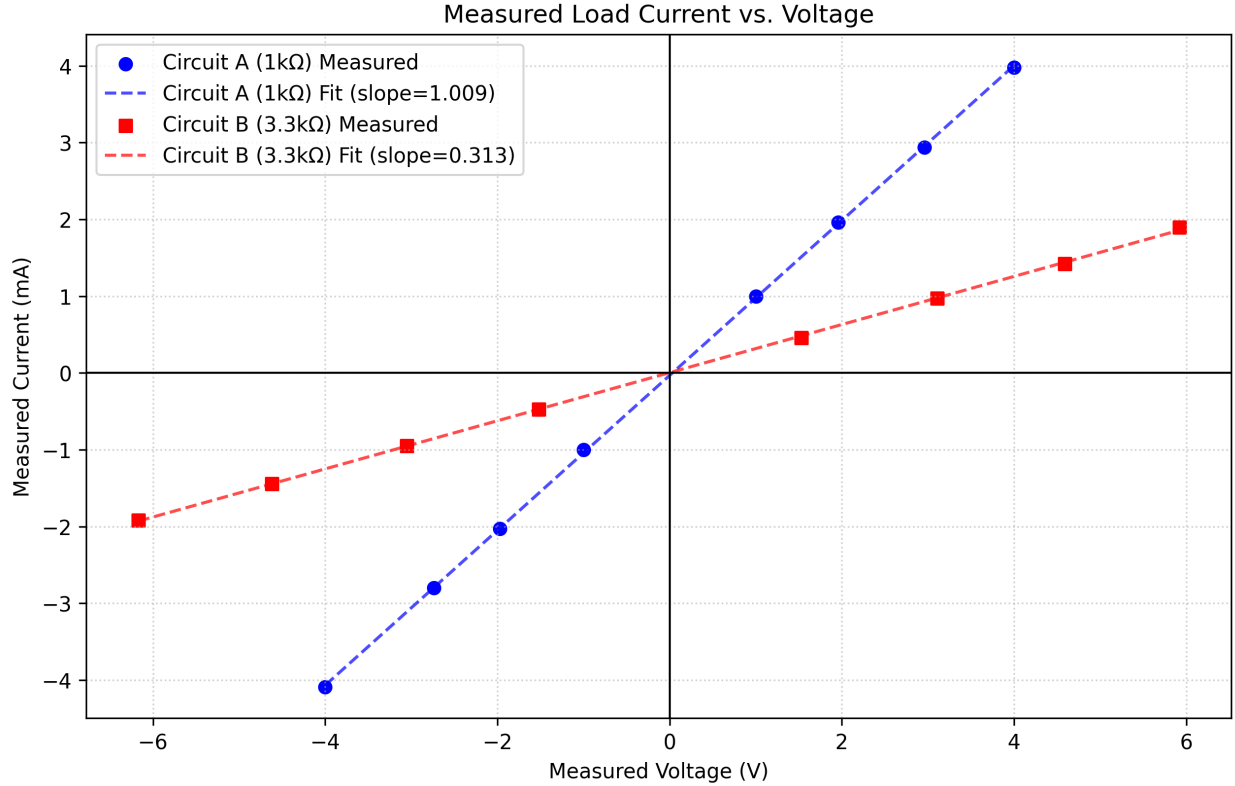


Figure 3: Comparison of measured current (mA) vs. voltage (V) for Circuit A ($R_2 = 1 \text{ k}\Omega$) and Circuit B ($R_2 = 3.3 \text{ k}\Omega$), showing linear Ohm's Law behavior and calculated best-fit lines.

V. Discussion, Analysis and Concluding Statements

A. Resistor Tolerances

All selected resistors ($1 \text{ k}\Omega$, $2.2 \text{ k}\Omega$, $3.3 \text{ k}\Omega$, $4.7 \text{ k}\Omega$, and $6.8 \text{ k}\Omega$) were found to be within their manufacturer stated tolerance ranges as seen in Table 1. The calculated percent error for all components remained low, with the highest error being 1.52% for the $3.3 \text{ k}\Omega$ resistor. This indicates that the resistors used in the experiment were of good quality and suitable for verifying Ohm's Law.

B. Verification of Ohm's Law

The relationship between current and voltage for both Circuit A and Circuit B was found to be linear, as shown in Figure 3.

- Circuit A Analysis:** For $R_2 = 1 \text{ k}\Omega$ in Table 2, the measured current and voltage closely matched the predicted values. For example, at $V_{in} = 8 \text{ V}$, the measured voltage across R_2 was 4.00 V , and the measured current was 3.98 mA , aligning with the predicted values.

- **Circuit B Analysis:** For $R_2 = 3.3\text{ k}\Omega$ in Table 3, the measured values followed the voltage divider and Ohm's Law predictions with similar consistency. At $V_{in} = 4\text{ V}$, the measured voltage was 3.11 V compared to the predicted 3.0698 V .

B. Slope and Significance

The slope of the current versus voltage plot in Figure 3 represents the conductance (G) of the load resistor, which is the reciprocal of the resistance ($G = 1/R$).

- In **Circuit A**, the best-fit slope was 1.009 , which corresponds to a resistance of approximately $1/1.009 \approx 0.991\text{ k}\Omega$.
- In **Circuit B**, the best-fit slope was 0.313 , corresponding to a resistance of approximately $1/0.313 \approx 3.195\text{ k}\Omega$.

These experimental slopes closely align with the measured resistance values recorded in Table 1, confirming the Ohm's Law relationship $I = V/R$.

C. Conclusion

This experiment demonstrated that the current through a resistor is directly proportional to the voltage across it. Small discrepancies between predicted and measured values can be attributed to the tolerances of the resistors and minor fluctuations in V_{in} from the DC power supply. Overall, the hypotheses were supported, and the validity of Ohm's Law was confirmed through the experimental data collected in both circuits.